LIFE SCIENCES · MERIDIAN LEARNING WORLD · BACKGROUND READING

The Same Four Moves

A deeper look behind Make and Break and Copy the Code: what the reactions really are, which pathways and cycles they run in — and why a goldfish, a shark, a pigeon and you, all built from the same chemistry, don't run it quite the same way. About a 15-minute read.

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PART 1 · THE MOVES, CLOSER UP

Make — condensation

CONDENSATION COSTS ENERGY



Every big molecule of life — protein, starch, glycogen, DNA, fat — is built by this one reaction, run with different building blocks. The detail the trainer hid: cells almost never condense *bare* monomers. They pre-charge them first, clipping each building block to an energy-rich carrier, and the "make" step then pays for itself when the carrier leaves:

- **Proteins:** each amino acid rides in on a tRNA, loaded at the cost of ATP → AMP + PP_i.
- **Glycogen:** glucose is first attached to a carrier called UDP; UDP-glucose is the real brick.
- **DNA:** the nucleotide arrives as a triphosphate (dNTP) — its own two spare phosphates are the ticket.
- **Fats:** fatty acids are activated as acyl-CoA — another ATP → AMP + PP_i "deep discharge".

The pattern: charge the brick, build with the charged brick, destroy the pyrophosphate so the wall can't un-build. You saw the ratchet in Copy the Code; it is everywhere.

Break — hydrolysis

HYDROLYSIS RELEASES ENERGY



The reverse move: water re-enters and the covalent bond splits. Digestion is hydrolysis running at industrial scale, with a dedicated pair of scissors for each bond type:

POLYMER	BOND CUT	SCISSORS (ENZYME)	WHERE	PRODUCTS
Starch, glycogen	glycosidic	amylase, maltase	mouth, small intestine	glucose
Proteins	peptide	pepsin, trypsin & co.	stomach, small intestine	amino acids
Fats	ester	lipase (+ bile to emulsify)	small intestine	fatty acids + glycerol
DNA, RNA	phosphodiester	nucleases	small intestine	nucleotides

Hydrolysis also runs inside every cell, quietly: worn-out proteins are shredded for parts, and proofreading polymerases hydrolyse their own fresh mistakes.

▼ Fine print · hydration is not hydrolysis

Sometimes water *joins* a molecule without cutting anything — it adds across a double bond and the chain stays whole. That is hydration, and the Krebs cycle has a famous one (fumarate + H_2O → malate). Hydrolysis breaks; hydration decorates. Examiners love the difference.

Stick — hydrogen bonds

NON-COVALENT REVERSIBLE

The δ+ ···· δ− lean that holds water together, zips DNA, and folds every protein into shape. One more job worth knowing: **recognition**. An enzyme grips its substrate — and a polymerase checks a base pair — largely through a handshake of hydrogen bonds. Right partner, all bonds line up; wrong partner, the handshake fails. Weak bonds are not a cheap substitute for real ones; they are the only bonds fast and forgiving enough to make and unmake a million times a second.

Pay — the ATP cycle

SPEND = HYDROLYSIS RECHARGE = CONDENSATION

$$ATP + H_2O \rightarrow ADP + P_i + \text{energy} \quad ADP + P_i + \text{energy} \rightarrow ATP + H_2O$$

Two ways the cell refills the battery. **Substrate-level:** a few reactions in glycolysis and the Krebs cycle hand a phosphate straight to ADP — quick, small change. **Oxidative phosphorylation:** the mitochondrion burns fuel to pump protons across its inner membrane, and the backflow spins ATP synthase — a literal rotary motor turning up to ~100 times a second, pressing ADP and P_i together. That machine makes ~90% of your ATP.

And the deep discharge: reactions that need an extra-hard shove cut ATP two notches down to AMP + PP_i (DNA ligase, amino-acid loading, fat activation). Free P_i never recharges AMP directly — adenylate kinase first rescues it: AMP + ATP → 2 ADP. Rising AMP is therefore the cell's sharpest low-battery alarm; a sensor protein (AMPK) reads it and switches the cell to power-saving mode.

The fifth move we skipped: pass the electrons

To keep the trainers honest-but-simple, one move stayed offstage: **redox** — transferring electrons (usually travelling as hydrogen) from one molecule to another. Burning is redox done all at once; respiration is the same chemistry in ~20 polite steps, with taxi molecules (NAD^+ → NADH) carrying electrons from food to the mitochondrial pumps. Most steps of the Krebs cycle and the entire electron-transport chain are this fifth move. Keep it in your pocket for the Energy unit.

PART 2 · WHERE THE MOVES LIVE: PATHWAYS AND CYCLES

A **pathway** is a production line (start material → product); a **cycle** is a line bent into a loop, where the last product regenerates the first ingredient. Spotting the four moves inside them turns a page of arrows into familiar chemistry:

PATHWAY / CYCLE	WHERE	THE MOVES AT WORK
Glycolysis (pathway)	cytoplasm, all cells	Spends 2 ATP to prime glucose, earns 4 back by substrate-level recharge; net +2 ATP, no oxygen needed.
Krebs cycle	mitochondrial matrix	Opens with a textbook **condensation** (acetyl-CoA + oxaloacetate → citrate), includes one hydration — and the rest is mostly the fifth move, loading NADH taxis.
β-oxidation (fat burning)	mitochondria	Snips fatty-acid chains two carbons at a time — a cousin of hydrolysis where CoA, not water, does the attacking — and feeds the pieces into Krebs.
Glycogen build / break	liver, muscle	Build: condensation from UDP-glucose. Break: not hydrolysis but **phosphorolysis** — phosphate attacks instead of water, so the freed glucose comes out pre-charged. Elegant frugality.
Urea cycle	liver	Condensations costing ~4 phosphate bonds per turn (one step is a true ATP → AMP + PP_i deep discharge), closed by a hydrolysis that snips urea off arginine.
Replication & protein synthesis	nucleus, ribosomes	Polymer factories: charged bricks, condensation 5′→3′ or N→C, PP_i ratchets, H-bond proofing — the whole of Copy the Code.
Calvin cycle (plants)	chloroplasts	The mirror image of respiration: light-charged ATP and electron taxis run condensations that build sugar from CO_2.

PART 3 · **SAME MOVES, DIFFERENT ANIMALS — AND WHY**

The core is untouchable: every animal runs the same four moves, the same ATP, the same genetic code. What evolution tunes is the **peripheral wiring** — and almost every difference traces back to one of four environmental pressures: water, diet, oxygen, temperature.

Water: three ways to dump nitrogen

Breaking surplus amino acids (hydrolysis, then deamination) leaves nitrogen as ammonia — cheap but toxic, and safe only when hugely diluted. What an animal does next depends on how much water it can spare:

WASTE	WHO	WATER COST	ATP COST	WHY IT FITS
Ammonia	most fish, aquatic larvae	huge	≈ none	Water everywhere — just let it diffuse out of the gills.
Urea	mammals, adult amphibians, sharks	moderate	~4 bonds per molecule (urea cycle)	Land life, limited water: pay ATP to package nitrogen into something storable. Sharks go further and deliberately *keep* urea in their blood as an osmotic ballast against seawater.
Uric acid	birds, reptiles, insects	minimal (white paste)	highest	Sealed eggs (waste must sit in the shell without poisoning the embryo), flight weight, deserts.

Same leftover nitrogen, three price plans — each trading ATP for water. Tadpole to frog, some amphibians literally switch plans mid-life as they leave the water.

Diet: pathways you can afford to lose

Vitamin C. Most animals build their own ascorbate from glucose — a short condensation-and-redox pathway. Our primate ancestors, living on fruit, lost the last enzyme (GULO) to mutation — molecular clocks put it near 60 million years ago — and never missed it — until sailors left the fruit behind and invented scurvy. Guinea pigs and some bats lost the same gene independently. The broken GULO still sits in your genome as a **pseudogene**: a molecular fossil, mutated exactly where the tree of life predicts.

Milk. All mammal infants make lactase, the scissors for milk sugar; most mammals — and most adult humans — switch the gene off after weaning. Several human populations that took up dairying (in Europe, East Africa, the Middle East) independently evolved regulatory mutations that keep lactase on for life. Different mutations, same effect, within the last ~10,000 years — evolution fast enough to carbon-date, and the reason lactose tolerance maps onto herding cultures.

Cellulose. Starch and cellulose are both pure glucose; the difference is one flip of the glycosidic bond's geometry (α vs β) — and no vertebrate makes scissors for the β version. Cows, sheep and termites eat grass and wood anyway, by subcontracting: fermenter chambers full of microbes that do hydrolyse cellulose, with the host living off the microbes' products. Your "fibre" is the cellulose your own enzymes must wave through.

Oxygen: fermentation with a twist

No oxygen, no electron-transport chain — then glycolysis alone must carry the battery load, and its waste becomes the problem. Your muscles briefly park the waste as lactate (the sprint burn). Crucian carp and goldfish, sealed under pond ice for months, cannot afford months of acid — so their muscles convert the waste onward to **ethanol**, which simply diffuses out through the gills. A fish that survives winter by brewing: same glycolysis as yours, one bolted-on enzyme step, driven by a very specific habitat.

Temperature: wasting energy on purpose

Mitochondria make ATP because protons must flow back through ATP synthase. Brown fat, abundant in hibernators and human babies, installs a bypass valve (the protein UCP1): protons leak straight through, no ATP is made, and the energy of food comes out as pure **heat**. Deliberately breaking the spend–recharge coupling from the Battery station is how a newborn — too small to shiver well — keeps warm, and how a hibernator reheats itself in spring. Pigs, oddly, carry a broken UCP1 gene, dead for ~20 million years — which is why piglets famously shiver and huddle in a pile.

▼ **Fine print · hydrogen bonds vs ice**

Antarctic fish swim in water slightly below 0 °C, kept liquid by salt. Their blood is less salty — it should freeze. It doesn't, because they evolved antifreeze glycoproteins: molecules that hydrogen-bond onto the faces of any nascent ice crystal and stop it growing. The Stick move, weaponised against winter — and in one fish lineage the gene demonstrably evolved from a digestive-enzyme gene. Evolution rarely invents; it repurposes.

PART 4 · **BACK TO THE TREE**

Notice what varied and what didn't. Nitrogen plans, vitamin pathways, antifreeze, ethanol tricks — all peripheral, all young, all explainable by habitat and diet. The four moves themselves, ATP, the code — identical from gut bacteria to goldfish to you,

unchanged since the last common ancestor. Biochemistry is a very old core wearing region-specific accessories; the accessories, including the broken genes, land on exactly the branches where the tree of life says they should. Which is, of course, how the tree was drawn: Build a Tree.

PART 5 · **READ THE ORIGINALS**

Nothing in this handout has to be taken on authority. These are the papers and lectures where the facts above were first established — most are free to read, several are short, and a few are very good stories. Original discovery papers are marked ●; readable retellings ○.

● **The citric acid cycle** — Krebs & Johnson, 1937. The journal *Nature* rejected the paper; it went on to become the most famous cycle in biology. Krebs tells the whole story himself in his Nobel lecture, 1953 (free PDF).

● **The urea cycle** — Krebs & Henseleit, 1932: the first metabolic cycle ever discovered, worked out on slices of liver. Original is in German; ○ a detailed retelling with the actual reasoning: ChemistryViews, "Sir Hans Adolf Krebs, Part 1".

● **How ATP is recharged (chemiosmosis)** — Peter Mitchell, 1961. The idea that a proton current powers ATP synthase was dismissed for years, then won the 1978 Nobel Prize in Chemistry. His Nobel lecture is the argument from the source (free PDF).

● **Watching the ATP motor spin** — Noji, Yasuda, Yoshida & Kinosita, *Nature* 386:299, 1997: they glued a fluorescent propeller to a single ATP synthase head and filmed the rotation under a microscope.

● **The broken vitamin-C gene** — Nishikimi & Udenfriend, *PNAS*, 1976 (no trace of the enzyme in primates or guinea pigs — free), then Nishikimi et al., *J. Biol. Chem.*, 1994: the dead GULO gene located and sequenced in the human genome.

● **Scurvy and the first controlled trial** — James Lind, 1753: twelve sailors, six treatments, citrus wins. The James Lind Library has the scanned pages of the experiment itself.

● **Lactase persistence, twice over** — Tishkoff et al., *Nature Genetics* 39:31, 2007: African and European dairying populations evolved different mutations with the same effect — convergent evolution recent enough to date.

● **The fish that brews** — ethanol as an anaerobic end-product was discovered by Shoubridge & Hochachka (*Science*, 1980); the enzyme duplication behind it: Fagernes et al., *Scientific Reports*, 2017 (free) — under winter ice, crucian carp run blood alcohol past the drink-driving limit.

● **Why piglets shiver** — Berg, Gustafson & Andersson, *PLoS Genetics*, 2006 (free): the pig lineage's UCP1 gene was disrupted ~20 million years ago.

● **Antifreeze from a digestive enzyme** — Chen, DeVries & Cheng, *PNAS* 94:3811, 1997 (free): the notothenioid antifreeze gene demonstrably evolved from trypsinogen — and the companion paper shows Arctic cod invented near-identical antifreeze independently.

○ **Three nitrogen plans compared** — Wright, *J. Exp. Biol.* 198:273, 1995: ammonia, urea and uric acid with their real water and energy price tags, across the animal kingdom.

U1.K2 · CONDENSATION & HYDROLYSIS

U5.K1/K3 · ATP, RESPIRATION & FERMENTATION

U6 · EVOLUTION & ADAPTATION

U8.K3 · HYDROGEN BONDS

Values and mechanisms follow standard references (Alberts et al., *Molecular Biology of the Cell*; Berg et al., *Biochemistry*). Companion trainers: Make and Break · Copy the Code · Build a Tree.